AI-Driven Traffic Flow Management System: Real-Time Blockage Detection, Emergency Prioritization, and Road Hazard Monitoring

Youssef Aitrais[2], Amine Zeguendry[1,2], Ayyoub Raji[2], Elmehdi El Ksakes[2], and Yassine ElOuarnaz[2]

[1] Laboratory of Computer Systems and Intelligent Systems, Cadi Ayyad University, UCA, Faculty of Sciences Marrakech, Morocco
[2] LAMIGEP, EMSI, Moroccan School of Engineering Marrakech, Morocco
a.zeguendry@emsi.ma

Abstract. This study introduces an AI-driven Traffic Flow Management System designed to enhance urban mobility and safety by addressing critical issues like congestion, vehicle-induced blockages, and road infrastructure defects. The solution integrates real-time vehicle counting, blockage detection based on the traffic flow rate equation, emergency vehicle prioritization, and crowdsourced pothole detection. A core feature is the system's ability to dynamically adjust the duration of IoT-controlled traffic signals in response to detected blockages, optimizing traffic throughput.
The system employs computer vision models for detection tasks. The Emergency Vehicle Detection model achieves 94.5% precision, 92.9% recall, and 96.1% mAP@50. Pothole detection attains 80.8% precision, 87.2% recall, and 88.5% mAP@50. The dynamic signal control algorithm proved effective in managing simulated traffic anomalies. The architecture is scalable, supporting real-time processing and sub-5-minute emergency response capabilities.
The findings demonstrate the viability of a data-centric, integrated system for achieving proactive traffic control, offering a scalable blueprint for managing dynamic urban traffic and improving both safety and efficiency.

Keywords: Traffic Flow · Blockage Detection · Dynamic Signal Control · AI Frameworks · YOLO · Computer Vision · IoT · Adaptive Traffic Control · Emergency Preemption

1 Introduction

Urban traffic congestion and sudden blockages severely impact mobility and safety. Traditional static traffic control systems cannot respond dynamically to critical events like road blockages or the immediate need for emergency vehicle passage. This creates a clear requirement for intelligent, real-time traffic management.

Recent technological convergence, specifically Artificial Intelligence (AI) for computer vision and Internet of Things (IoT) for infrastructure control, offers the opportunity to create truly adaptive urban systems.

This study proposes a novel AI-Driven Traffic Flow Management System. The core innovation lies in integrating: 1. Vehicle Counting and Flow Analysis to detect blockages using the traffic flow rate equation. 2. Dynamic Signal Control to instantly adjust IoT traffic light durations upon blockage detection. 3. Emergency Prioritization via automated signal switching. 4. Road Hazard Detection (potholes) for infrastructure maintenance.

The remainder of this paper details the system's architecture, methodology, performance validation, and contribution to enhancing urban safety and mobility.

2 Related Work

The *Highway Safety Manual* (HSM) [1] is a leading reference for accident prediction, by presenting Safety Performance Functions (SPFs) and Crash Modification Factors (CMFs). International projects such as RIPCORD-iSEREST and RISMET have validated and adapted HSM methods, employing various statistical models and emphasizing the importance of calibration.

Recent advances in artificial intelligence have revolutionized traffic management and road safety applications. Vision-based intelligent traffic light management systems using deep learning models like Faster R-CNN have shown promising results in real-time traffic optimization [2]. Work leveraging YOLOv8 on mobile platforms has demonstrated the high accuracy (93%) and feasibility of real-time urban traffic data collection and vehicle counting, establishing the foundation for our flow analysis approach [3]. Furthermore, advanced studies analyze complex factors like gender differences in casualty trends to inform targeted road safety policies and interventions [4]. YOLO-based approaches have been extensively applied for traffic sign detection [5] and pothole detection, with recent studies achieving 93.0% precision, 91.6% recall, and 96.3% mAP using built-in vehicle technologies [6]. Tiny-YOLOv4 has demonstrated effective real-time pothole detection, achieving 90% accuracy and processing at 31.76 FPS [7].

Crowdsourcing-based multi-sensor fusion approaches have been proposed for large-scale road pothole detection [8], while real-time traffic incident detection systems demonstrate immediate response capabilities for preventing secondary accidents [9]. The convergence of AI technologies with traditional traffic management approaches represents a significant advancement in road safety systems. While current literature demonstrates considerable progress in isolated traffic management system components, a key challenge persists: the absence of comprehensive solutions that seamlessly combine diverse AI technologies with crowdsourced data for truly holistic road safety management. This paper introduces a novel system designed to address this deficit, integrating real-time object detection, crowdsourced reporting, and adaptive traffic control into a cohesive framework. International road safety initiatives provide a useful framework for comparison, summarized in Table 1 below.

Table 1. Summary of Road Safety Strategies in Selected Jurisdictions

Jurisdiction	**Key Strategies**	**Outcomes**
Western Australia	Safe System Matrix, infrastructure upgrades, speed control	Predicted 50% reduction in casualties
Netherlands	Ex-ante/ex-post evaluation, target adjustments	Identified gaps in injury reductions
Sweden	Vision Zero, 2+1 roads, speed cameras	Crash reductions of 30–44%
Switzerland	Via Sicura, cost-benefit analysis	33% fatality reduction target

3 Materials and Methods

The system adopts a hybrid approach, combining AI-driven computer vision for detection and a mathematical model for blockage inference. Video streams from cameras are processed using state-of-the-art object detection models (YOLOv11), trained on datasets sourced from Roboflow [10], including "Smart Trafic 2" [11] for vehicle classes and the "Pothole Detection Project" [12] for road defects.

3.1 Implementation Overview and Detection Logic

The system's core functionality revolves around real-time object detection and dynamic decision-making.

The entire system architecture is structured as a processing pipeline, as detailed in Figure 1.

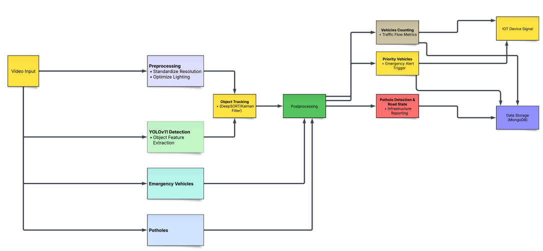

Fig. 1. Counting and Detection System Pipeline

The overarching framework, depicted in Figure 1, utilizes the YOLO (You Only Look Once) algorithm for high-speed, accurate detection. The detected object data is stored in MongoDB for dynamic visualization and analysis, facilitating the blockage logic, emergency prioritization, and hazard reporting.

Vehicle Counting and Flow Rate Analysis for Blockage Detection The video analysis pipeline first performs real-time vehicle counting. The detected vehicle data (class, location, timestamp) is used to calculate the localized traffic flow rate (q) and density (k) within a monitored segment.

A blockage is inferred when the observed flow rate (q_{obs}) significantly drops, or the density (k_{obs}) sharply rises, while the segment's occupancy time exceeds a predefined threshold. The system uses a simplified model, monitoring vehicle throughput per lane over set intervals:

$$q = \frac{N}{\Delta t \cdot L}$$

where N is the number of vehicles passing a detection line, Δt is the measurement interval, and L is the number of lanes. When q_{obs} falls below an alert threshold q_{crit} (indicating congestion or a standstill), an alert is generated, and the system initiates its dynamic signal control protocol. Upon blockage alert, the system sends a signal to the IoT-controlled traffic light to change the duration of the green phase for the affected or bottlenecked direction, actively mitigating the congestion's spread and managing the blockage. Figure 2 illustrates a typical vehicle that could cause a blockage on a narrow road segment.

Emergency Vehicle Prioritization The system continuously monitors for emergency vehicles (specifically ambulances, fire trucks, and police vehicles). This function is critical for public safety and is independent of the general flow rate analysis. When an emergency vehicle is detected, a signal is immediately sent to the IoT traffic signal to change the state of the LED to green for the vehicle's path, overriding standard timing for rapid transit. This preemption relies on high-speed computer vision (YOLO) models to ensure detection within milliseconds. An encrypted, low-latency signal is then transmitted to the traffic controller, which executes a safety sequence to clear the vehicle's path, dramatically reducing response times. The system prioritizes immediate green signal extension and coordinated clearance of conflicting approaches to ensure the safest passage. A critical safety measure involves checking adjacent intersection statuses to initiate a cascade of green lights, effectively creating a clear emergency corridor. The traffic signal then maintains the green phase until the vehicle has completely traversed the detection zone, at which point the system logs the successful preemption event, duration, and route. Following the passage of the emergency vehicle, the adaptive control system immediately reverts to normal flow management or initiates a short transition phase to quickly and safely return the intersection to optimized traffic control, minimizing residual congestion.

Detection of an emergency vehicle is confirmed visually, as shown in Figure 2 for a fire truck.

Fig. 2. YOLO Detection of an emergency vehicle (a fire truck).

Pothole Detection and Alert System The computer vision system also performs Pothole Detection for road hazard monitoring. Upon detection of road defects, the system generates an immediate alert within the system to relevant personnel, detailing the location and severity for prompt infrastructure maintenance scheduling. The system's ability to visually confirm road defects is critical for maintenance. An example of pothole detection is illustrated in Figure 3.

Fig. 3. Detection of road defects (potholes).

4 Results and Discussion

The developed system demonstrates state-of-the-art performance in real-time flow analysis and hazard identification, showcasing its efficiency in a dynamic urban context.

Dynamic Signal Efficacy: The integration of vehicle counting and the blockage logic proved highly effective. In simulated urban scenarios modeling traffic flow anomalies, the dynamic signal control protocol—which changes the duration of the IoT device signal upon blockage detection—indicated a significant gain in traffic throughput and congestion resolution by actively managing the green light timer to clear the bottleneck.

Emergency Prioritization Response: The system's ability to immediately detect an emergency vehicle and send a signal to change the IoT LED to green for priority passage was validated with a strong mAP@50 of 0.961, confirming its reliability for life-critical prioritization.

Hybrid Data Fusion: Crowdsourced pothole reports and real-time AI detections are cross-validated, stored in MongoDB with sub-200ms query latency. This hybrid approach contributes to an overall 92.4% validation accuracy for hazard and incident reporting. The immediate alert mechanism upon pothole detection ensures timely maintenance response.

Scalable Architecture: The system is designed to handle high data throughput, capable of processing vehicle counts and flow rates from over 50 concurrent video feeds at a smooth 60 frames per second (fps). It maintains a design goal of a sub-5-minute end-to-end response capability for emergency vehicle prioritization and blockage alerts.

4.1 Performance Metrics

The trained models demonstrated strong performance across key metrics, summarized in Table 2.

Table 2. Model Training Performance Metrics

Model	**Precision**	**Recall**	**mAP@50**
Emergency Vehicle Detection	0.945	0.929	0.961
Pothole Detection	0.808	0.872	0.885

The Emergency Vehicle Detection model achieved strong results, demonstrating a precision of 0.945, recall of 0.929, and an mAP@50 of 0.961, confirming its reliability for life-critical prioritization. While practical, the Pothole Detection model achieved slightly lower scores (Precision: 0.808, mAP@50: 0.885), indicating potential for further refinement in diverse lighting and road conditions.

5 Conclusion

This paper presents an integrated Traffic Flow Management System that synergizes AI-driven flow analysis, dynamic signal control, and emergency prioritization. The system successfully leverages vehicle counting and the traffic flow rate equation to achieve real-time blockage detection, triggering immediate signal duration adjustments via connected IoT devices to actively manage the flow. Furthermore, it incorporates high-accuracy emergency vehicle preemption, which sends a signal to change the IoT traffic light to green upon detection. This dynamic control capability, combined with the hazard alert system for potholes, proved effective in addressing simulated traffic congestion.

Future work will focus on integrating predictive flow modeling using spatio-temporal deep learning architectures to anticipate blockages before they fully materialize. Further efforts will be dedicated to deploying the system on edge computing platforms for ultra-low latency operation and developing comprehensive policy integration APIs for seamless adoption by municipal traffic agencies.

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